# Pinellas County Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 09 April, 2026

## Contents

# Indicators

## Nutrients

### Total Nitrogen - Discrete

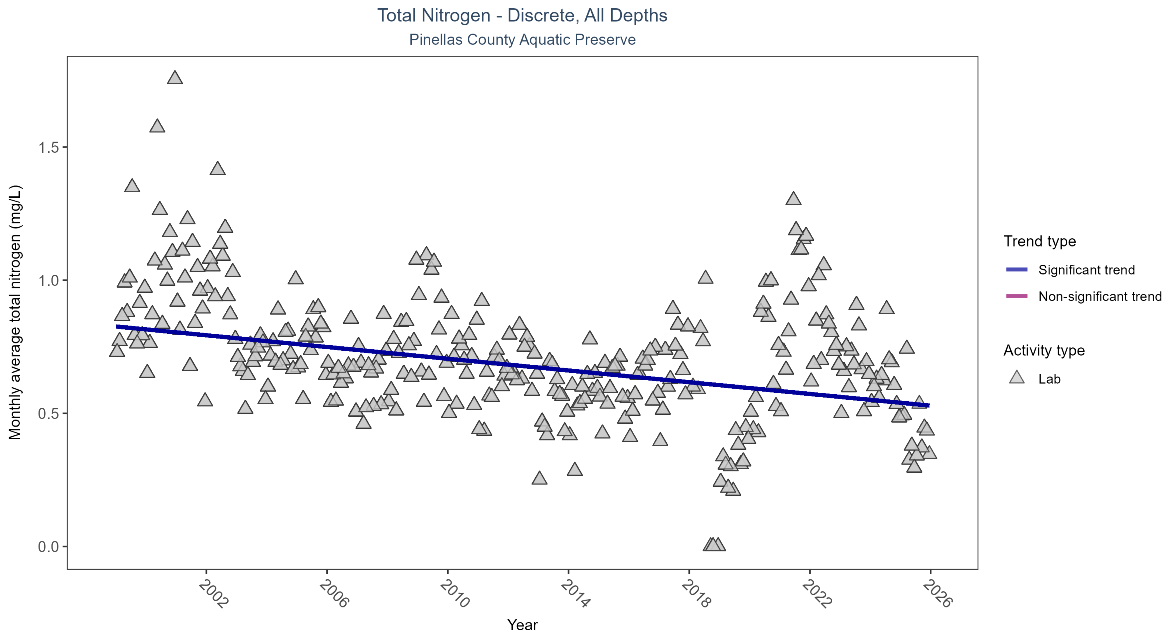


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Seasonal Kendall-Tau Results for - Total Nitrogen

| Activity Type | Statistical Trend | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Lab | Significantly decreasing trend | 16755 | 27 | 1999 - 2025 | 0.589 | -0.29071 | 0.82623 | -0.01102 | 0 |

Monthly average total nitrogen decreased by 0.01 mg/L per year.

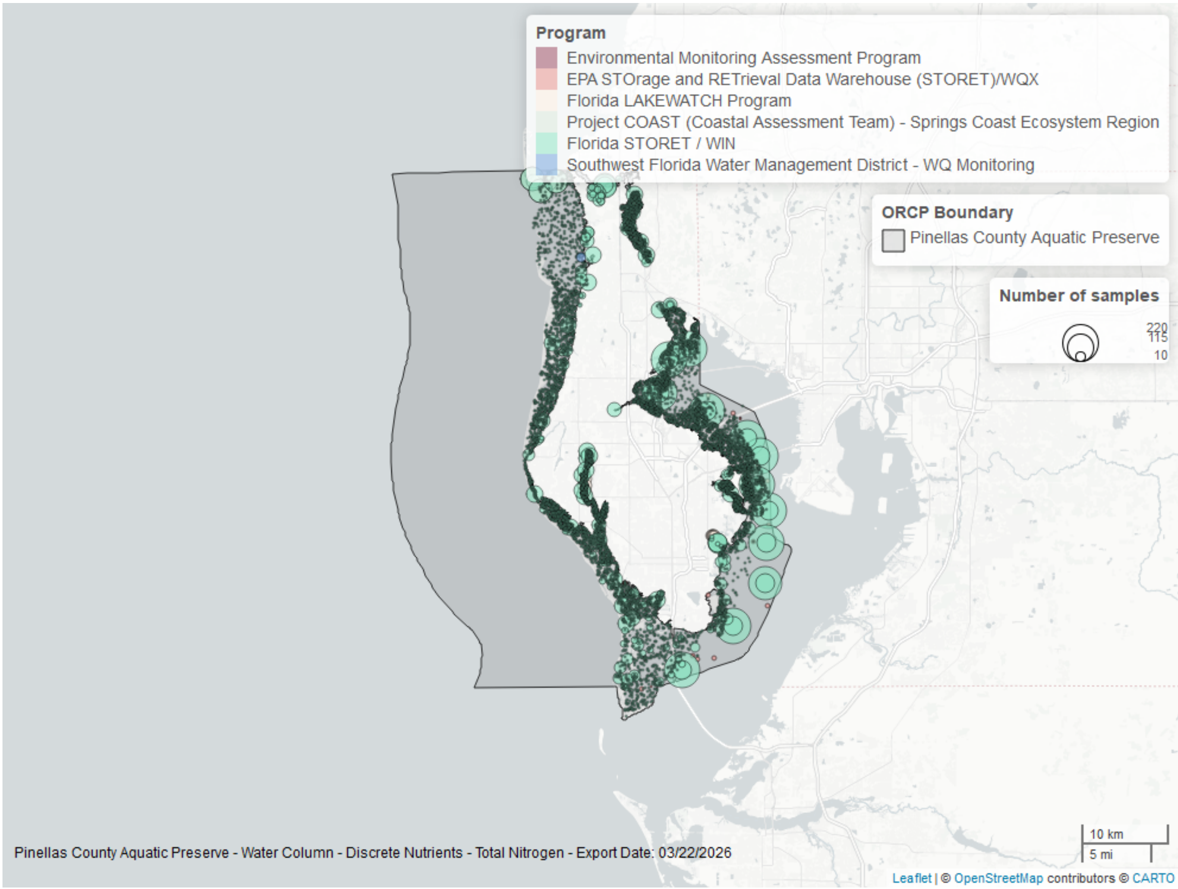


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Total Phosphorus - Discrete

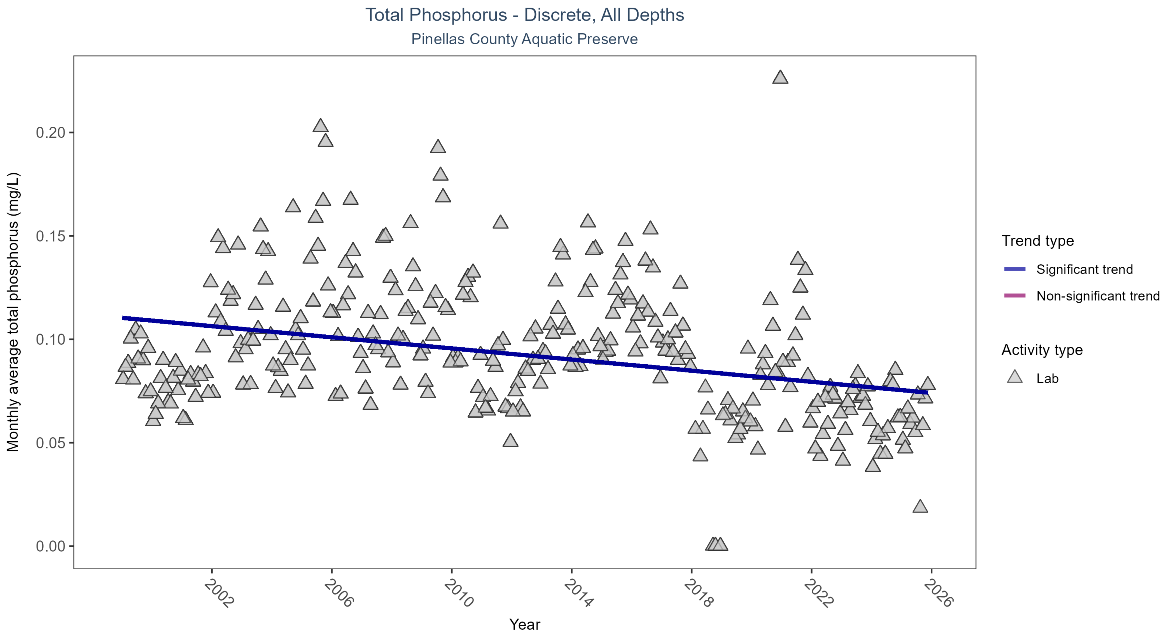


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Seasonal Kendall-Tau Results for - Total Phosphorus

| Activity Type | Statistical Trend | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Lab | Significantly decreasing trend | 16551 | 27 | 1999 - 2025 | 0.087 | -0.27304 | 0.11039 | -0.00134 | 0 |

Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

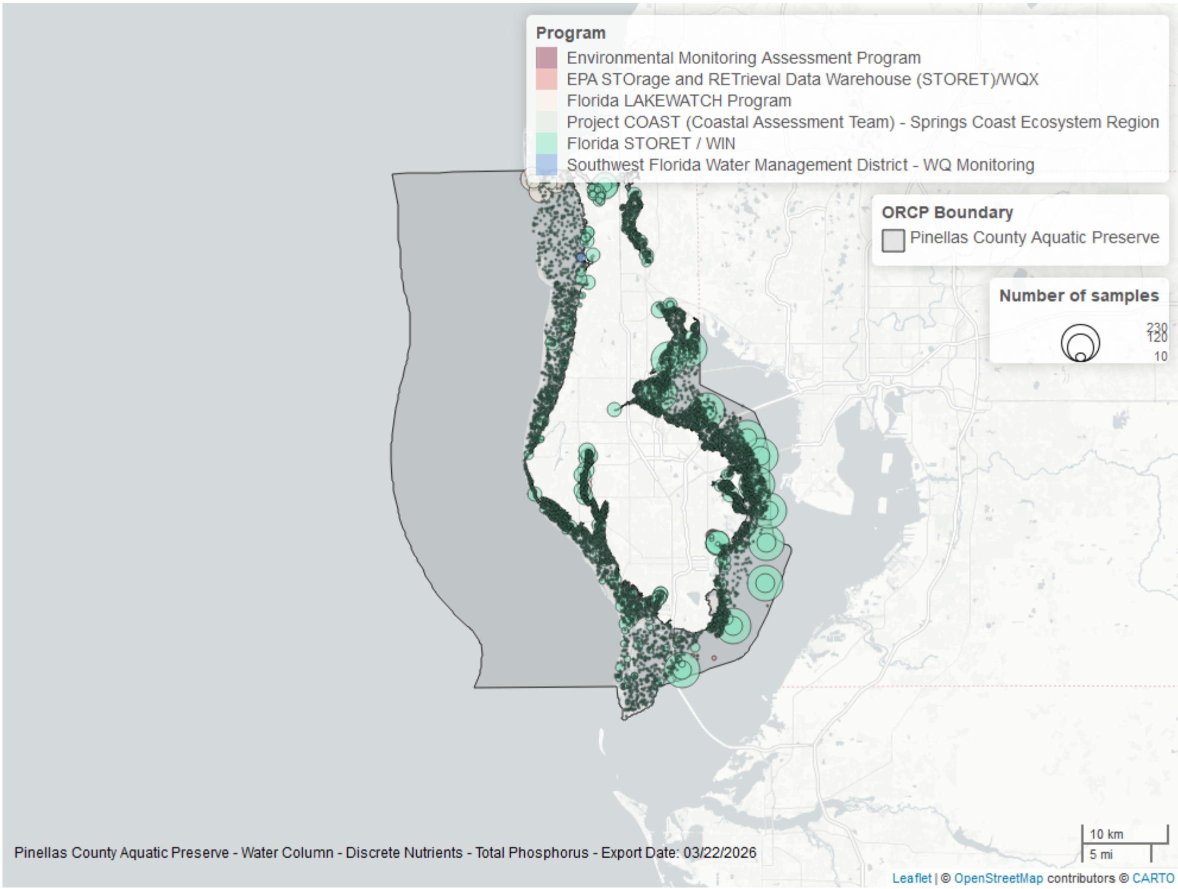


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

# Water Quality

## Dissolved Oxygen - Discrete

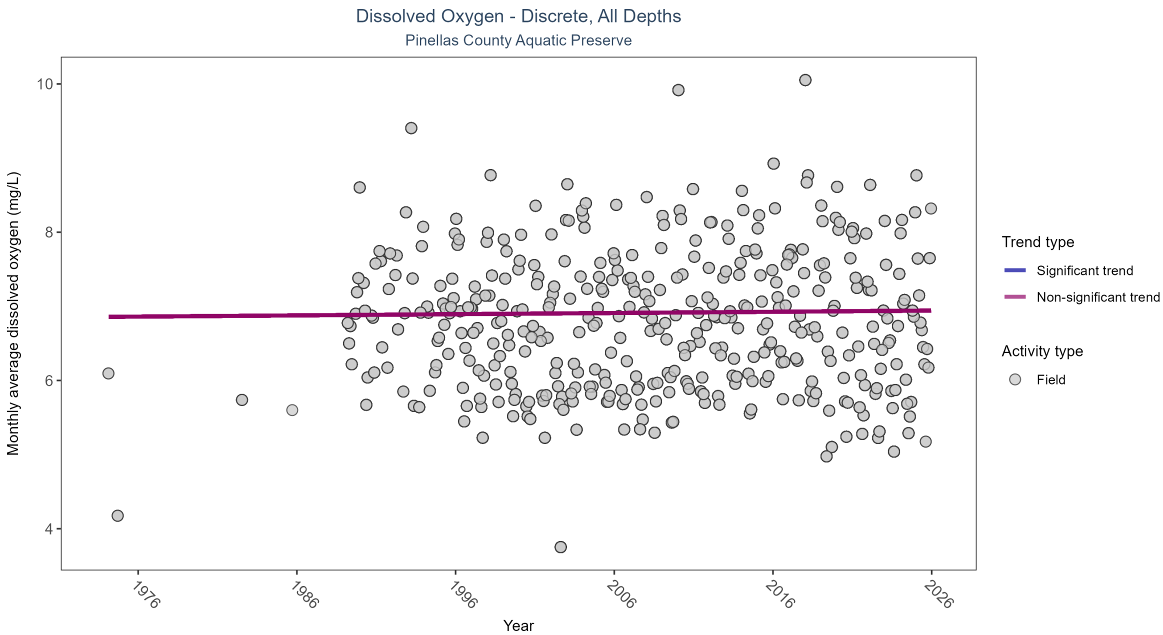


Figure 5: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 3: Seasonal Kendall-Tau Results for - Dissolved Oxygen

| Activity Type | Statistical Trend | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Field | No significant trend | 97306 | 40 | 1974 - 2025 | 6.6 | 0.02966 | 6.85817 | 0.00163 | 0.4634 |

Dissolved oxygen showed no detectable trend between 1974 and 2025.

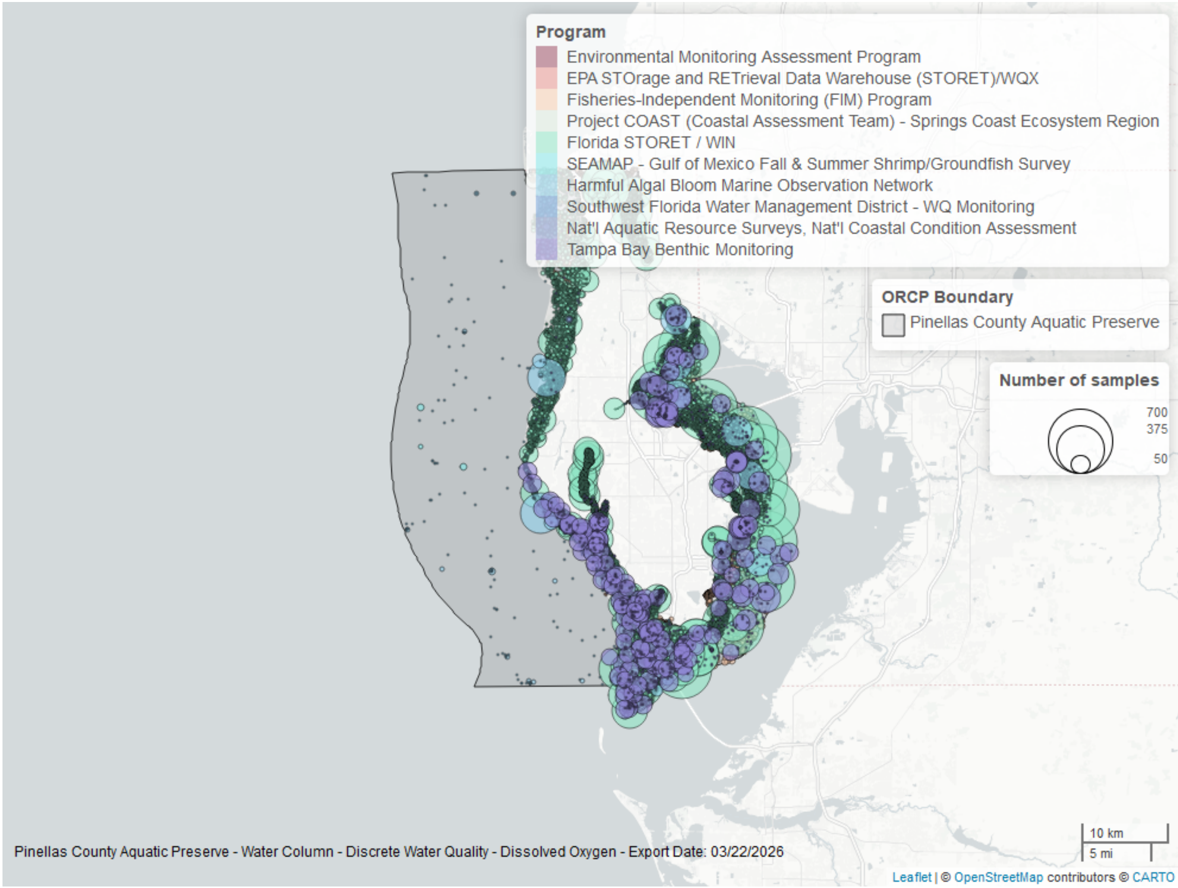


Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

### Dissolved Oxygen Saturation - Discrete

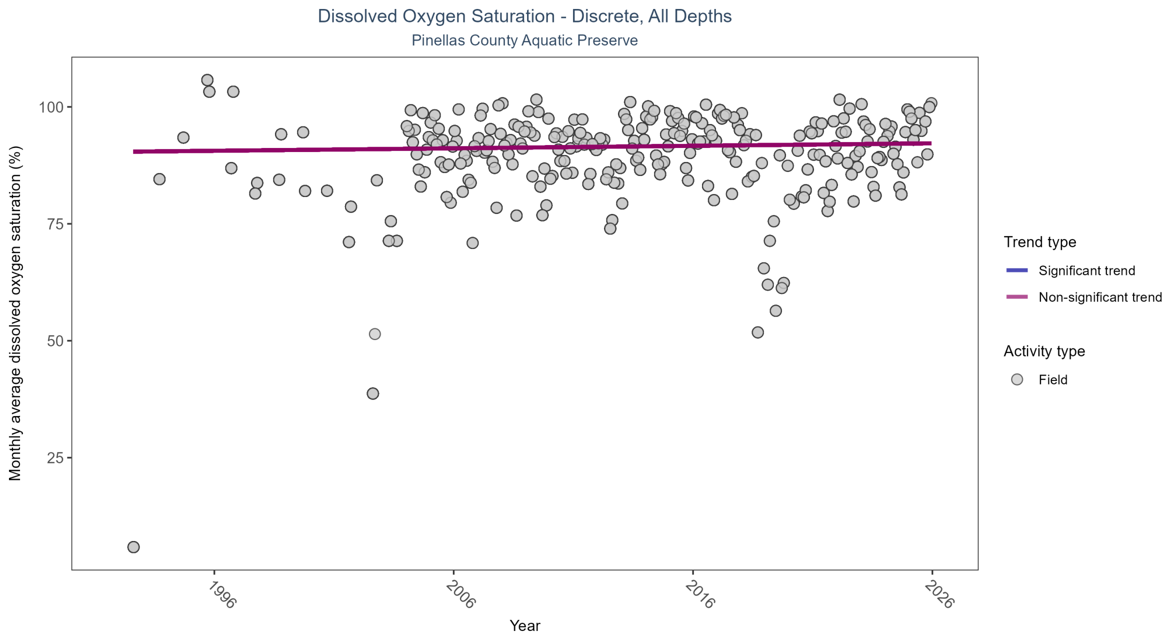


Figure 7: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 4: Seasonal Kendall-Tau Results for - Dissolved Oxygen Saturation

| Activity Type | Statistical Trend | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Field | No significant trend | 31862 | 34 | 1992 - 2025 | 91 | 0.05016 | 90.4014 | 0.05317 | 0.319 |

Dissolved oxygen saturation showed no detectable trend between 1992 and 2025.

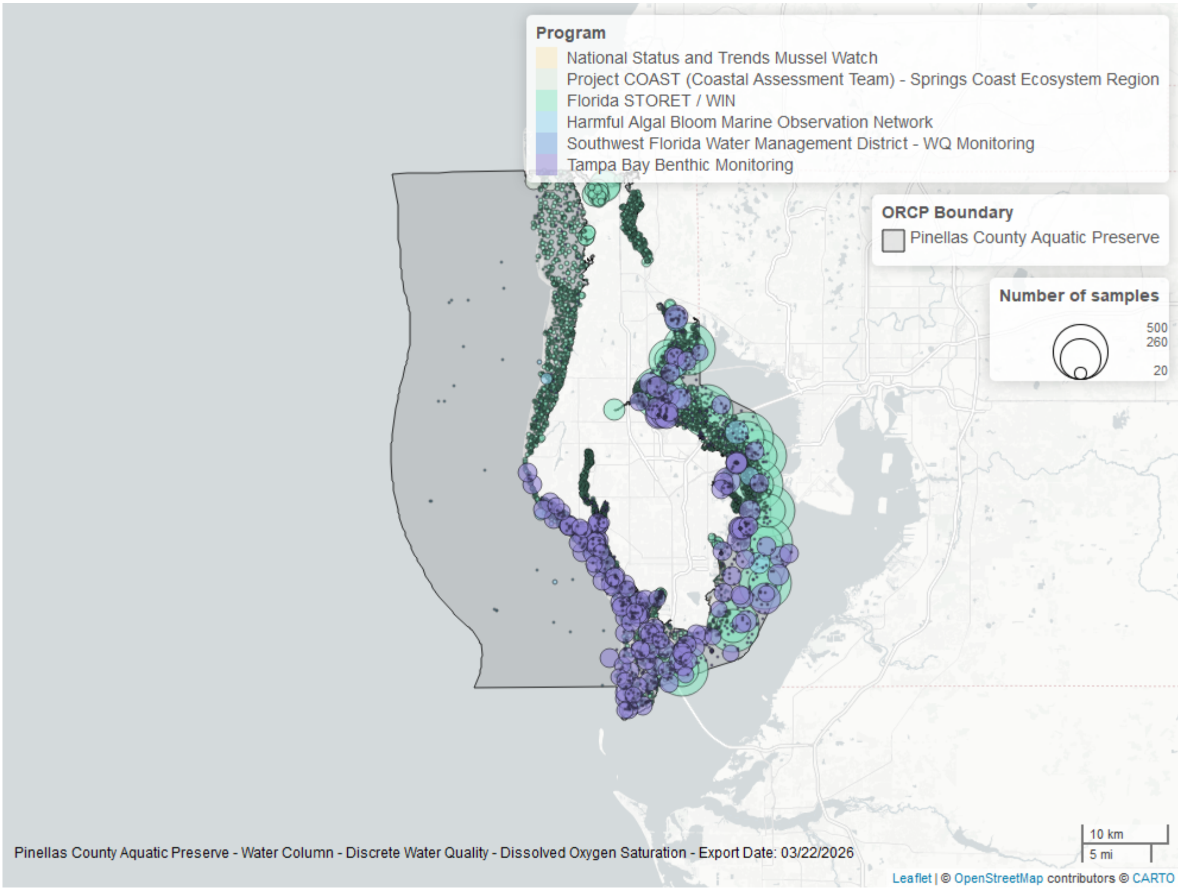


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Salinity - Discrete

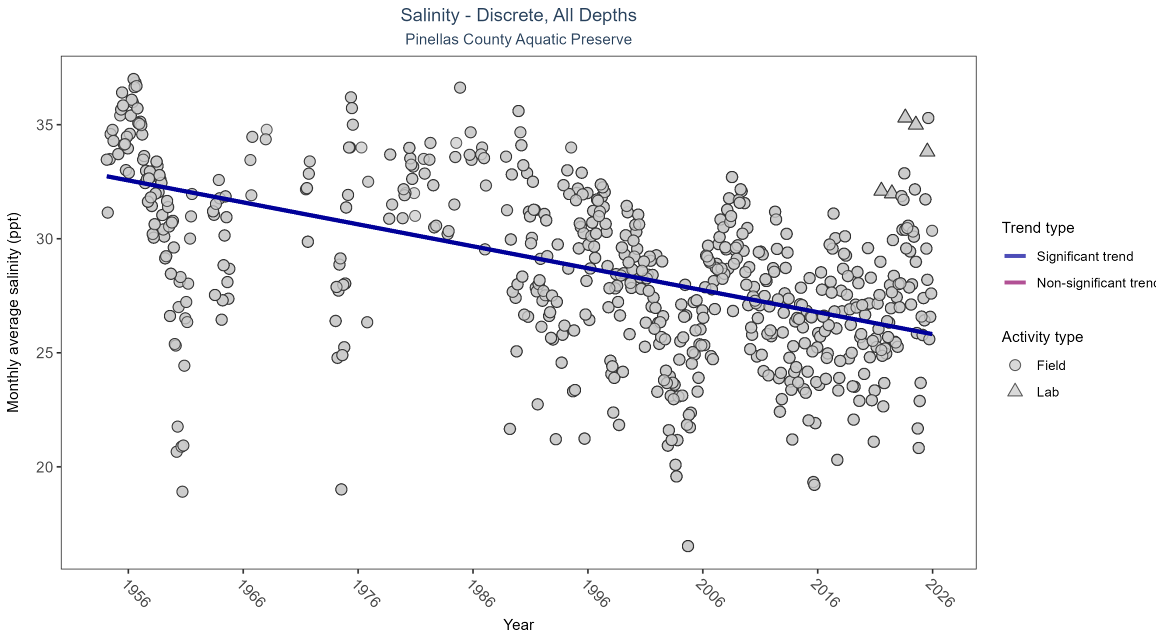


Figure 9: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 5: Seasonal Kendall-Tau Results for - Salinity

| Activity Type | Statistical Trend | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| All | Significantly decreasing trend | 96840 | 65 | 1954 - 2025 | 28.96 | -0.36067 | 32.75008 | -0.09631 | 0 |

Monthly average salinity decreased by 0.1 ppt per year.

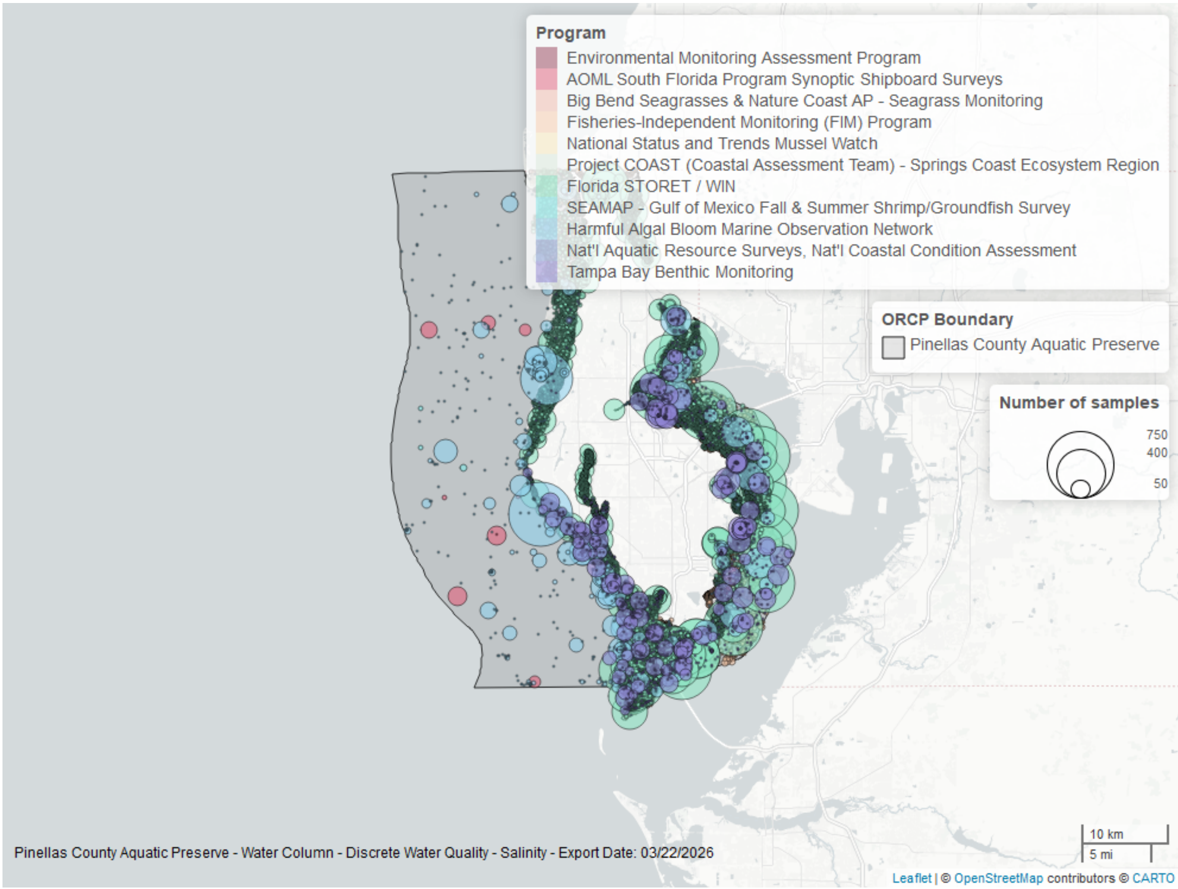


Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Water Temperature - Continuous

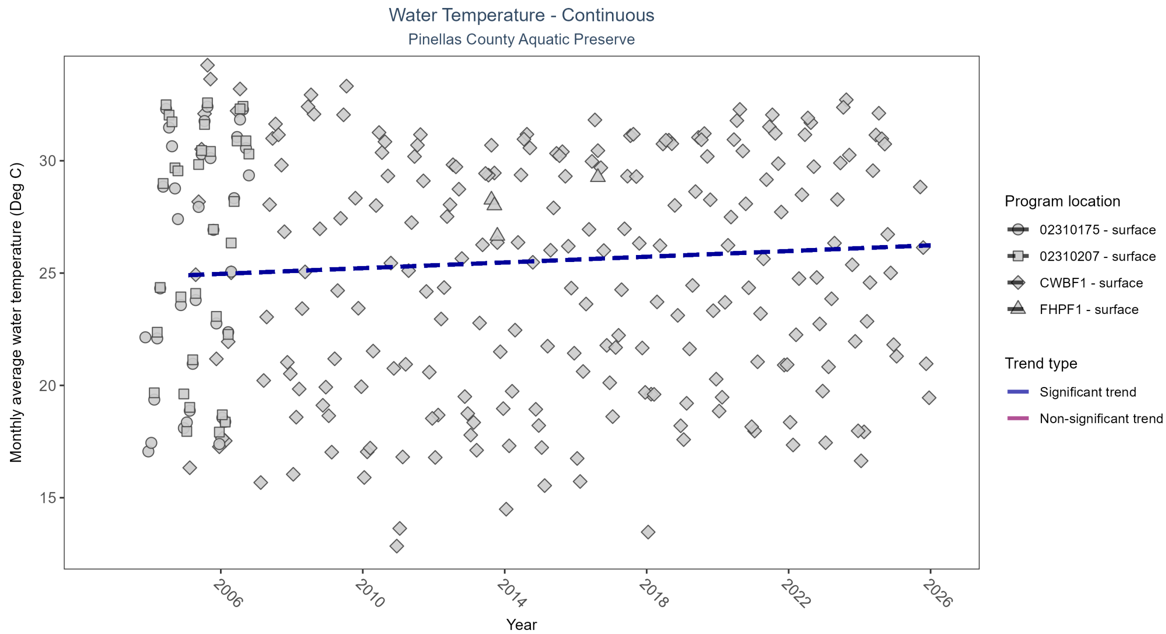


Figure 11: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 6: Seasonal Kendall-Tau Results - Water Temperature

| Program Location | Statistical Trend | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| CWBF1 | Significantly increasing trend | 1443248 | 21 | 2005 - 2025 | 25.40 | 0.18 | 24.9 | 0.06 | 0.0001 |
| FHPF1 | Insufficient data to calculate trend | 12636 | 2 | 2013 - 2016 | 27.90 | - | - | - | - |
| 02310175 | Insufficient data to calculate trend | 1421 | 4 | 2003 - 2006 | 26.50 | - | - | - | - |
| 02310207 | Insufficient data to calculate trend | 1424 | 3 | 2004 - 2006 | 27.55 | - | - | - | - |

At one program location, monthly average water temperature increased by 0.06°C per year. There was insufficient data to fit a model for three locations.

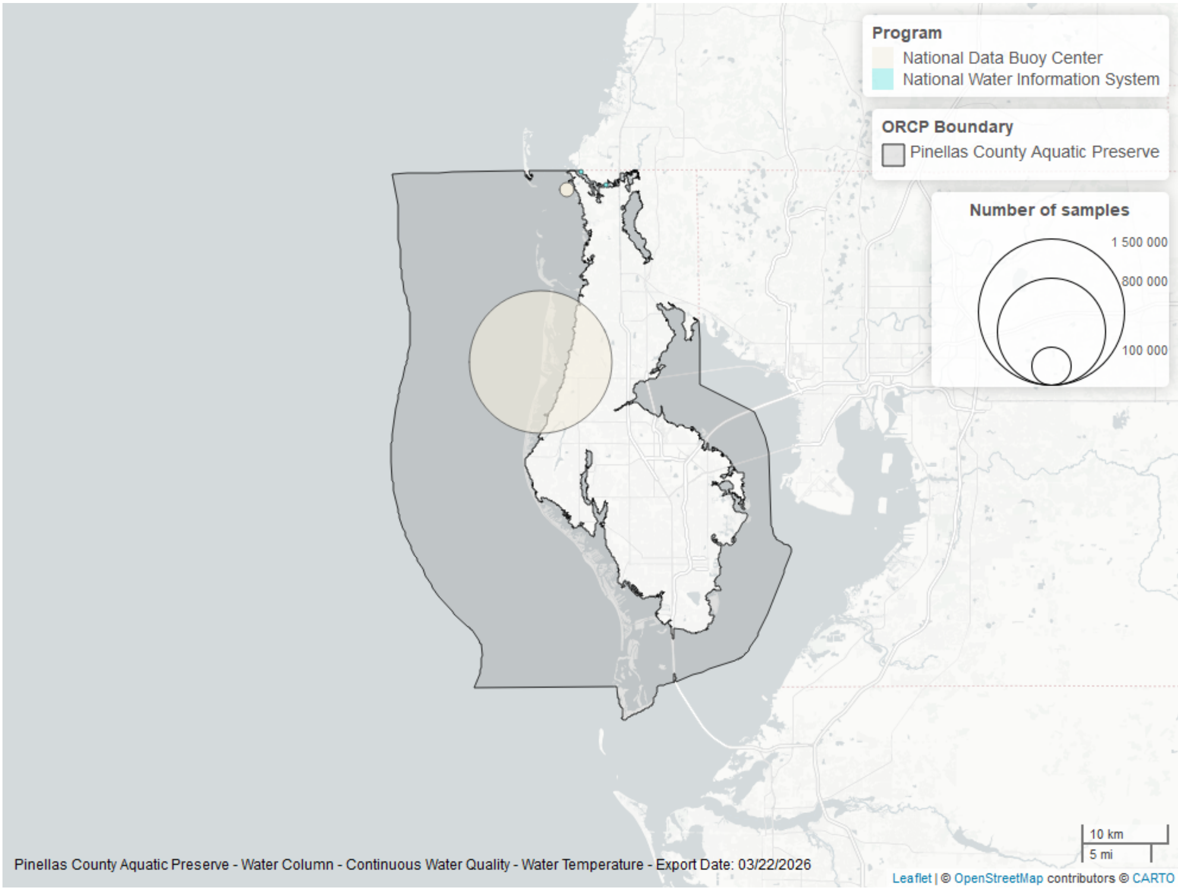


Figure 12: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

### Water Temperature - Discrete

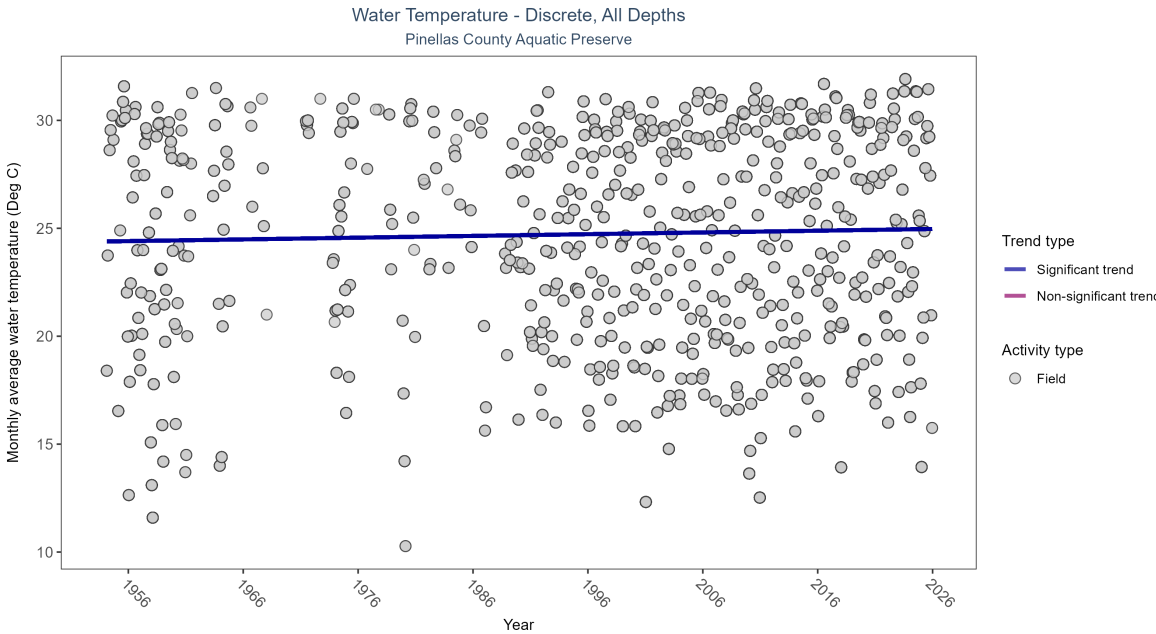


Figure 13: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 7: Seasonal Kendall-Tau Results for - Water Temperature

| Activity Type | Statistical Trend | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Field | Significantly increasing trend | 103247 | 68 | 1954 - 2025 | 26.55 | 0.08979 | 24.38917 | 0.00805 | 0.0016 |

Monthly average water temperature increased by 0.01°C per year.

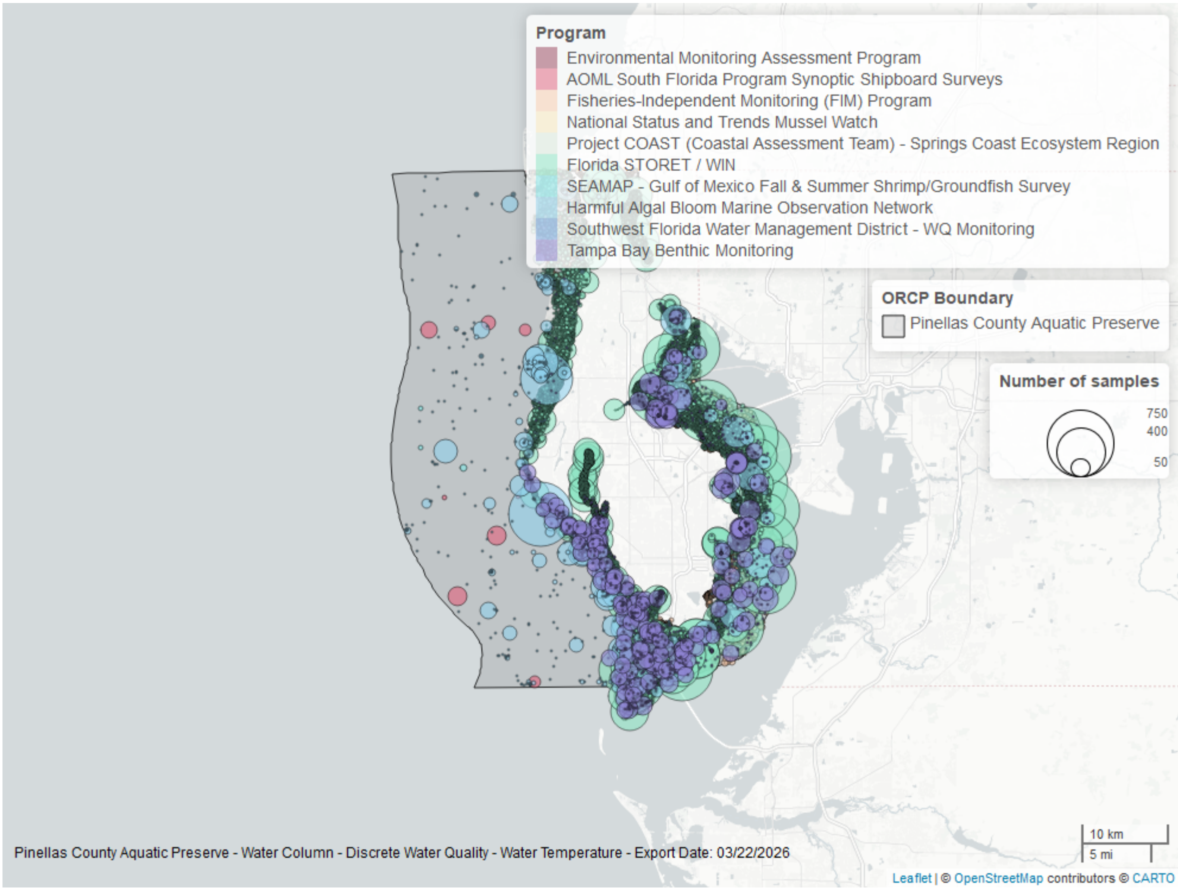


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

### pH - Discrete

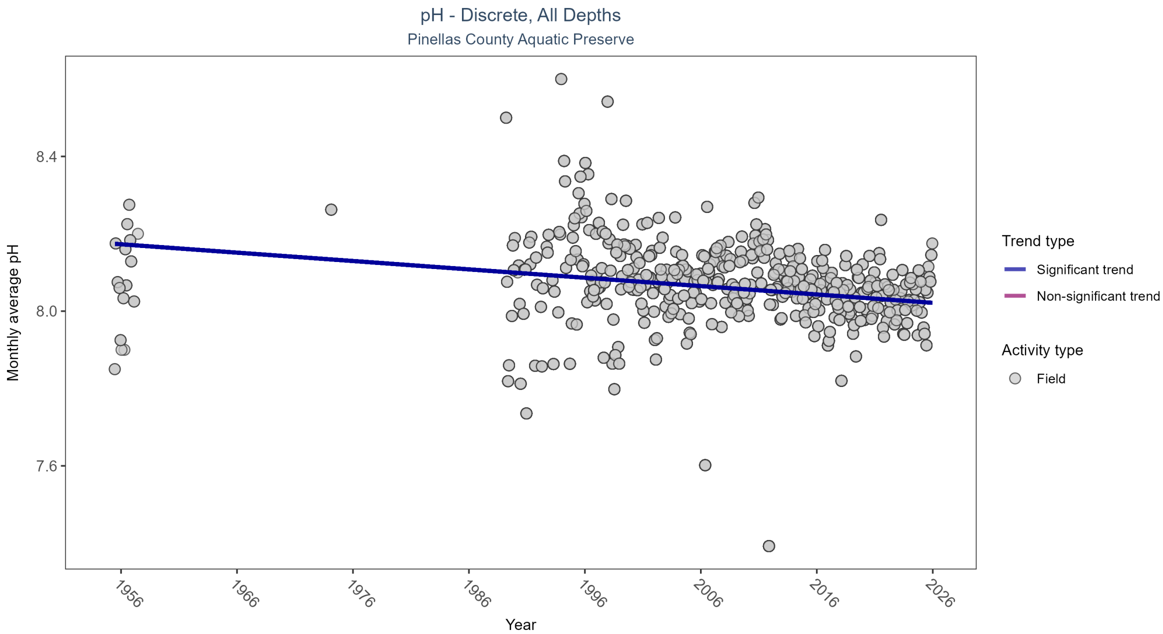


Figure 15: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 8: Seasonal Kendall-Tau Results for - pH

| Activity Type | Statistical Trend | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Field | Significantly decreasing trend | 92528 | 41 | 1955 - 2025 | 8.1 | -0.2069 | 8.17504 | -0.00216 | 0 |

Monthly average pH decreased by less than 0.01 pH units per year.

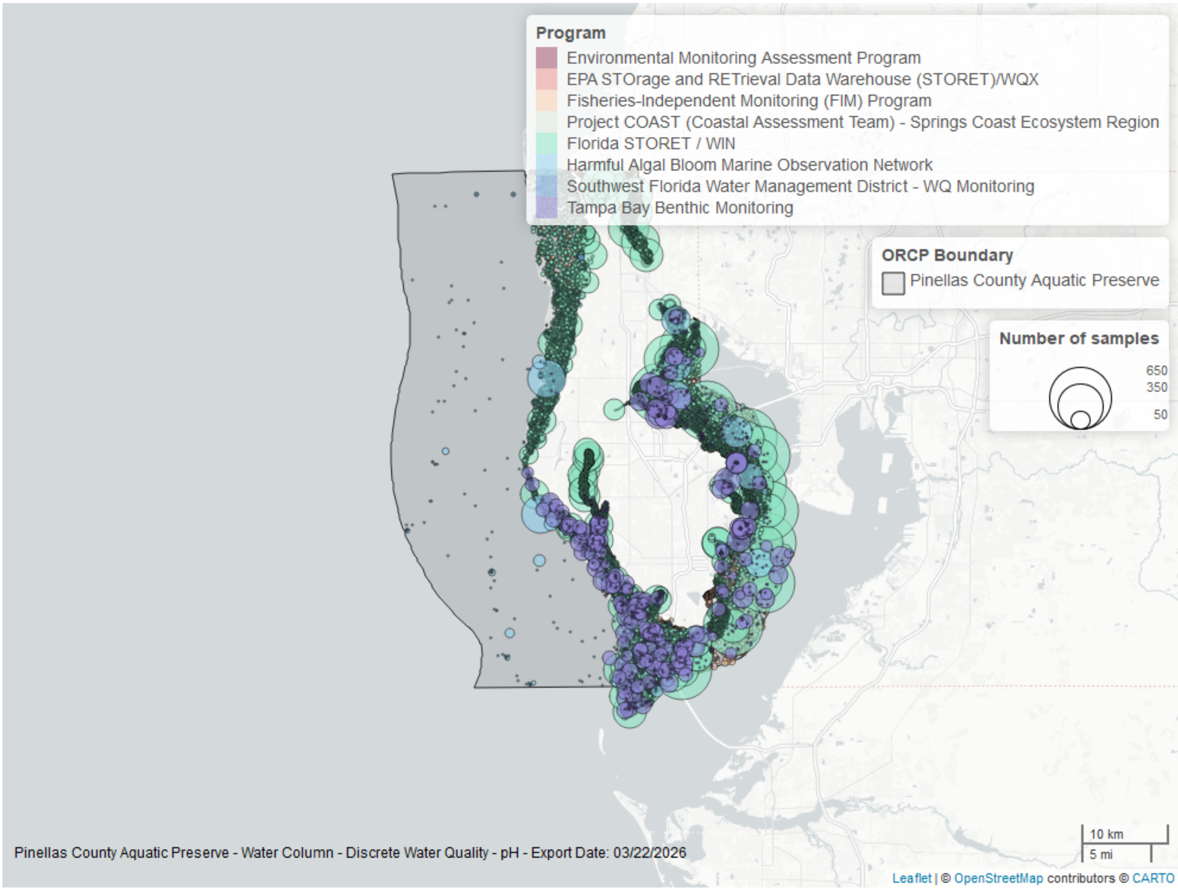


Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Specific Conductivity - Continuous

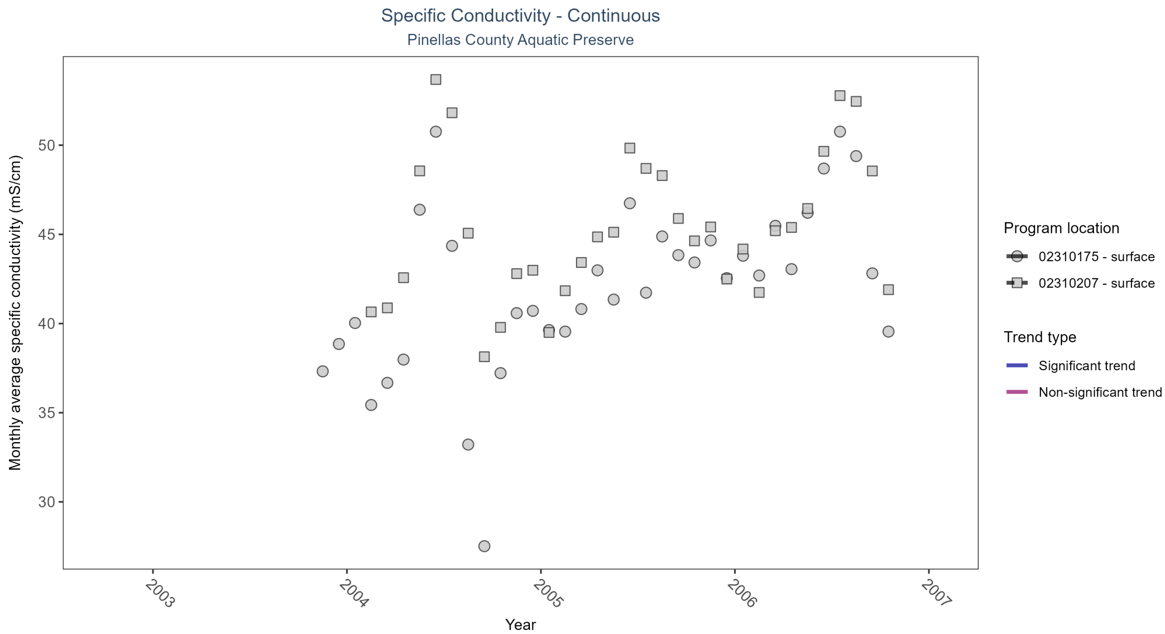


Table 9: Seasonal Kendall-Tau Results - Specific Conductivity

| Program Location | Statistical Trend | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| 02310175 | Insufficient data to calculate trend | 1876 | 4 | 2003 - 2006 | 42.3 | - | - | - | - |
| 02310207 | Insufficient data to calculate trend | 1502 | 3 | 2004 - 2006 | 45.6 | - | - | - | - |

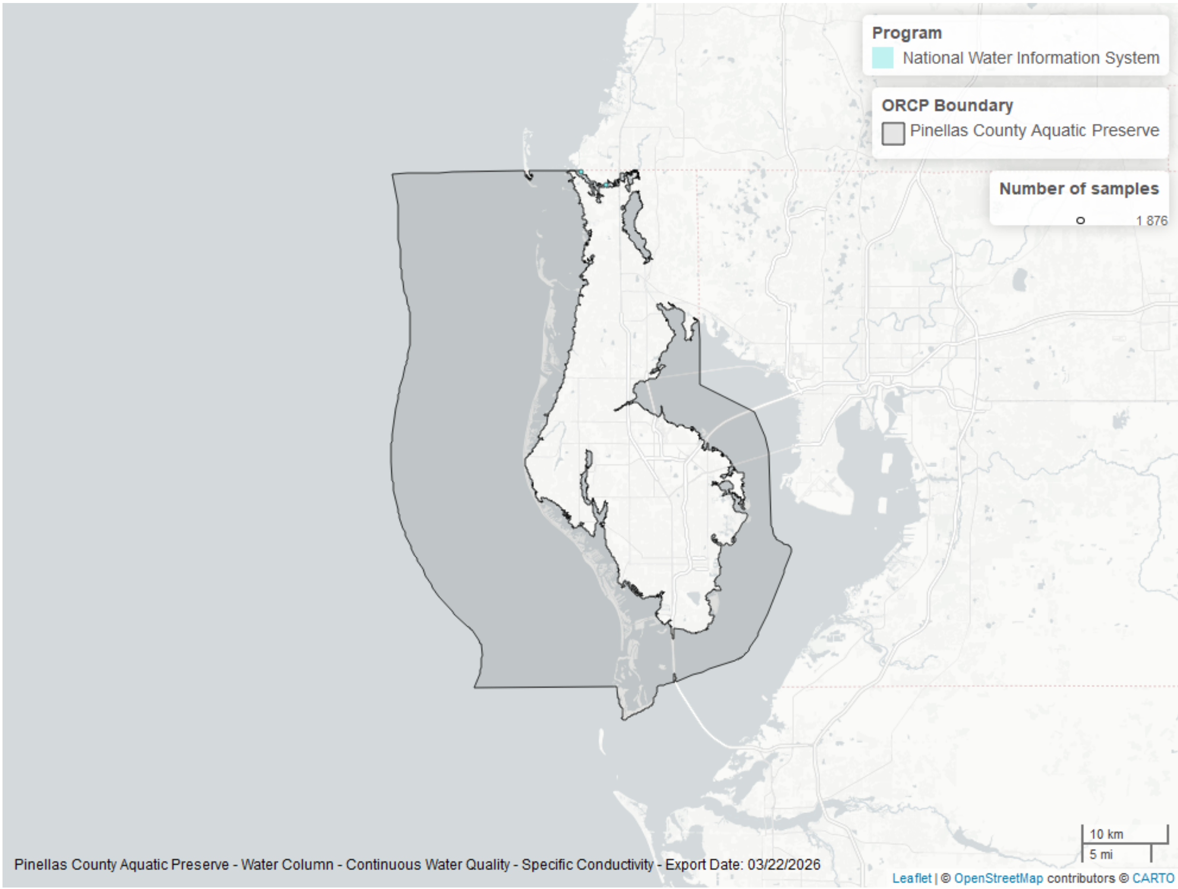


Figure 17: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

# Water Clarity

## Turbidity - Discrete

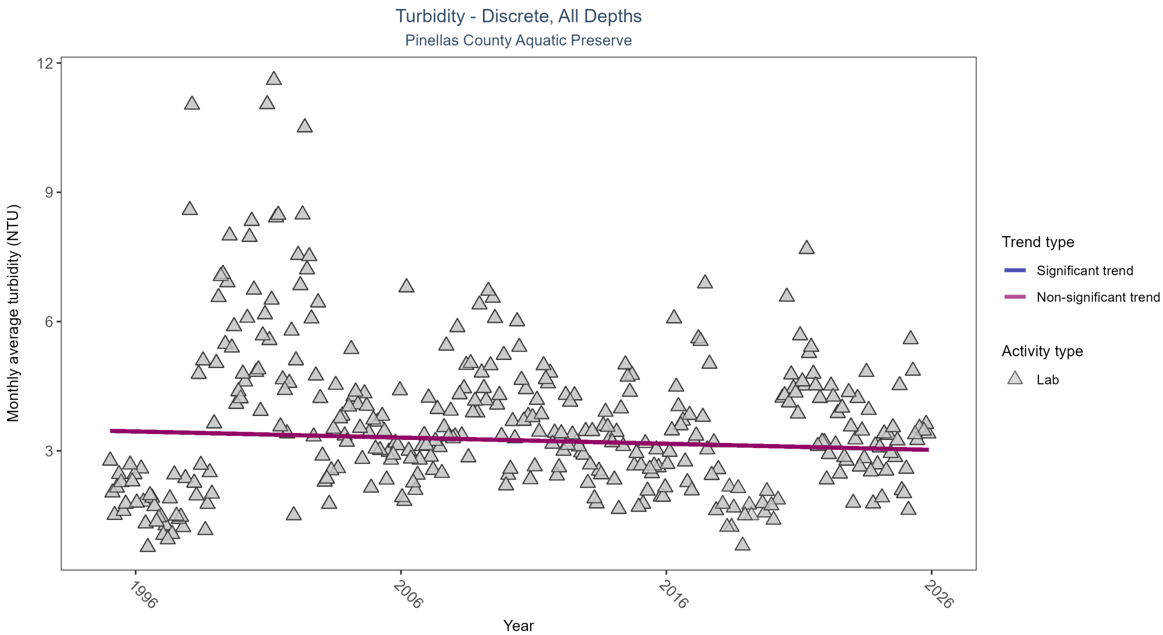


Figure 18: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 10: Seasonal Kendall-Tau Results for - Turbidity

| Activity Type | Statistical Trend | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Lab | No significant trend | 21801 | 31 | 1995 - 2025 | 2.5 | -0.05191 | 3.46372 | -0.01424 | 0.1751 |

Turbidity showed no detectable trend between 1995 and 2025.

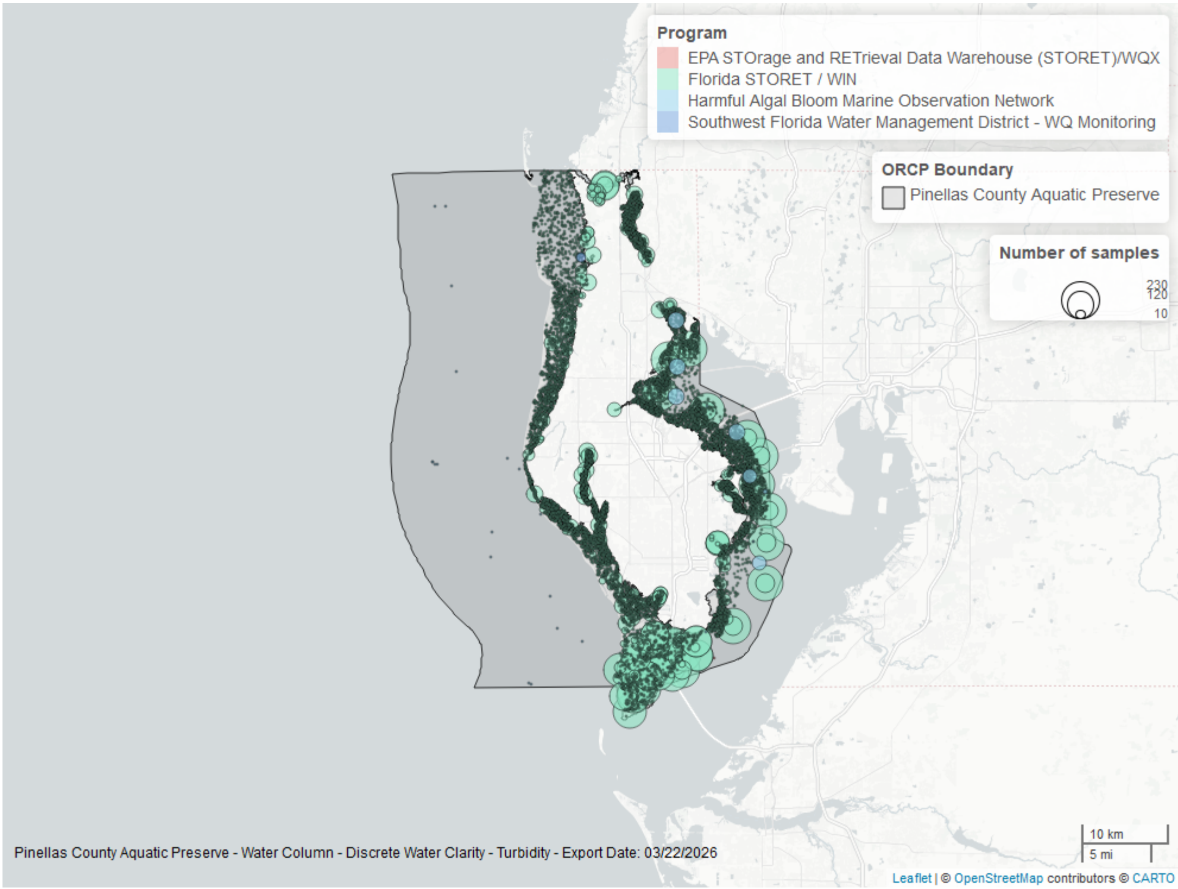


Figure 19: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Total Suspended Solids - Discrete

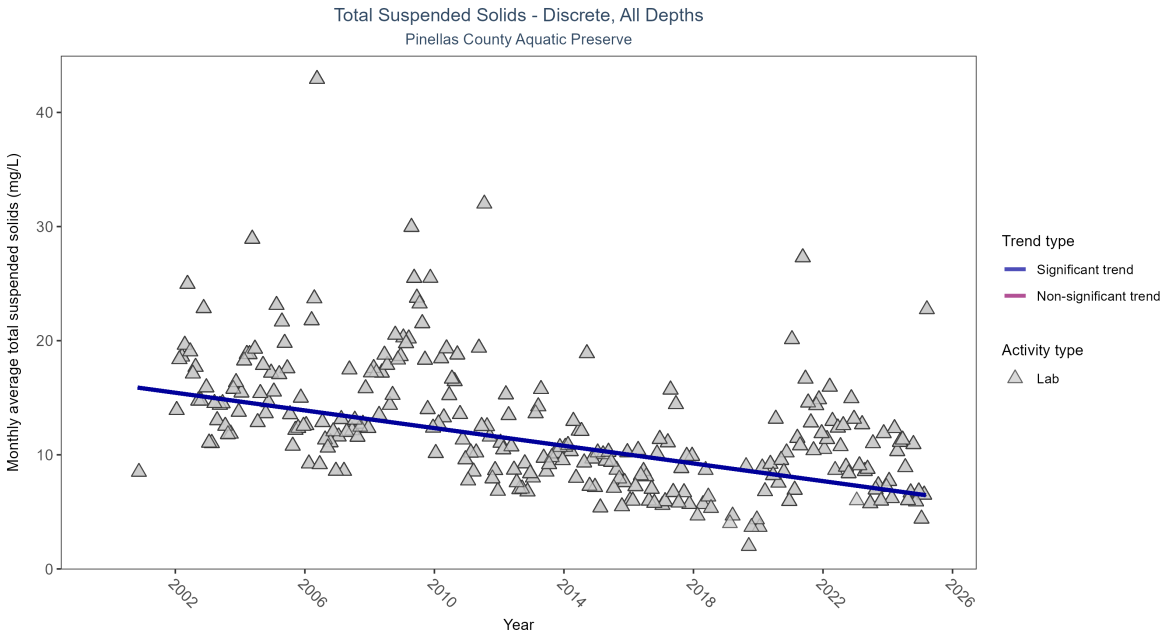


Figure 20: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 11: Seasonal Kendall-Tau Results for - Total Suspended Solids

| Activity Type | Statistical Trend | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Lab | Significantly decreasing trend | 12668 | 25 | 2000 - 2025 | 10 | -0.40465 | 16.21286 | -0.38724 | 0 |

Monthly average total suspended solids decreased by 0.39 mg/L per year, indicating an increase in water clarity.

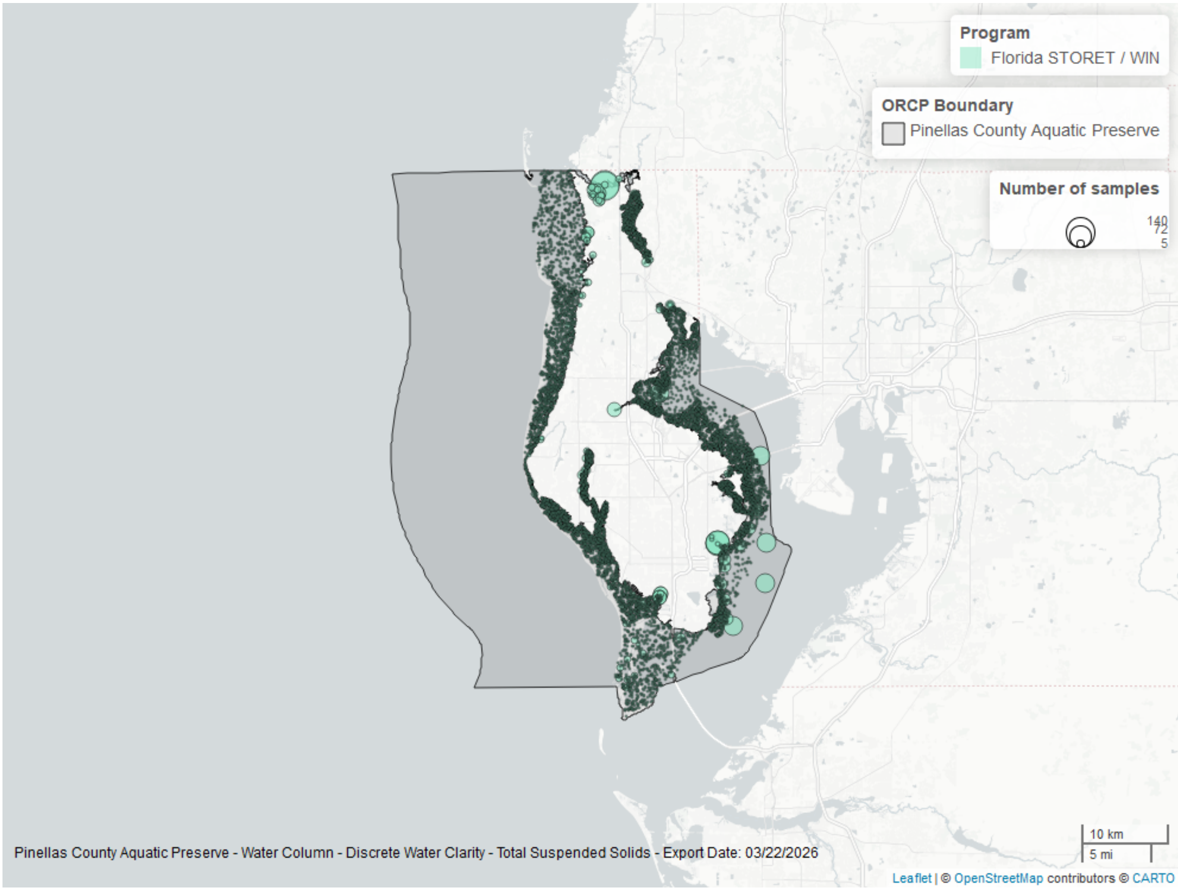


Figure 21: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Chlorophyll a, Uncorrected for Pheophytin - Discrete

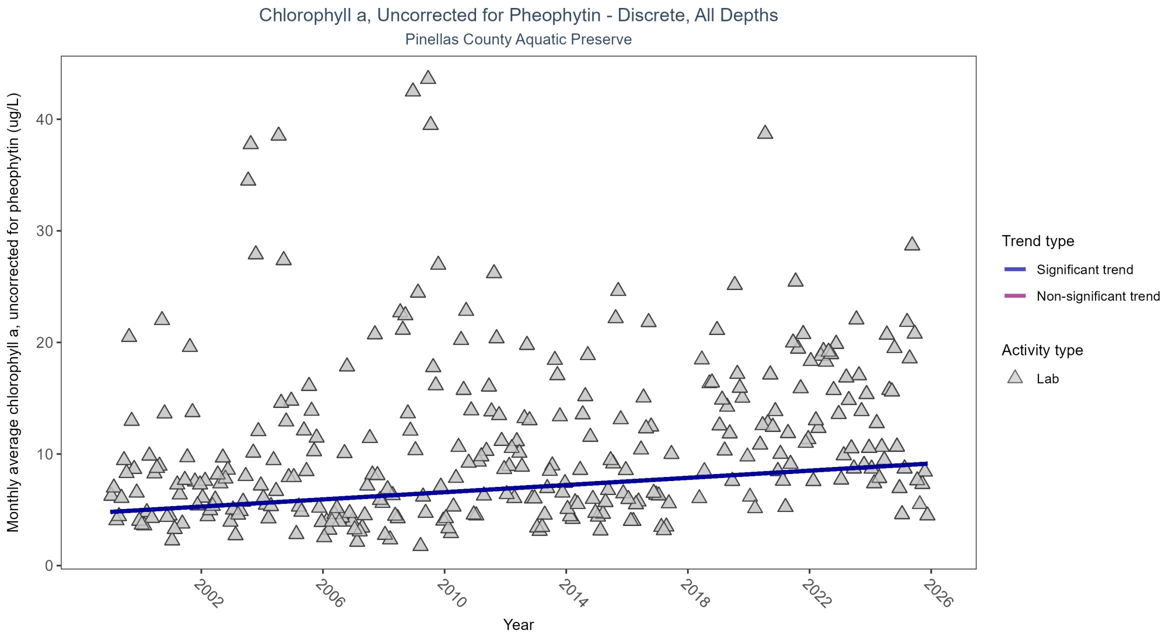


Figure 22: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 12: Seasonal Kendall-Tau Results for - Chlorophyll a, Uncorrected for Pheophytin

| Activity Type | Statistical Trend | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Lab | Significantly increasing trend | 7148 | 27 | 1999 - 2025 | 6.4 | 0.24102 | 4.80601 | 0.16093 | 0 |

Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.16 µg/L per year, indicating a decrease in water clarity.

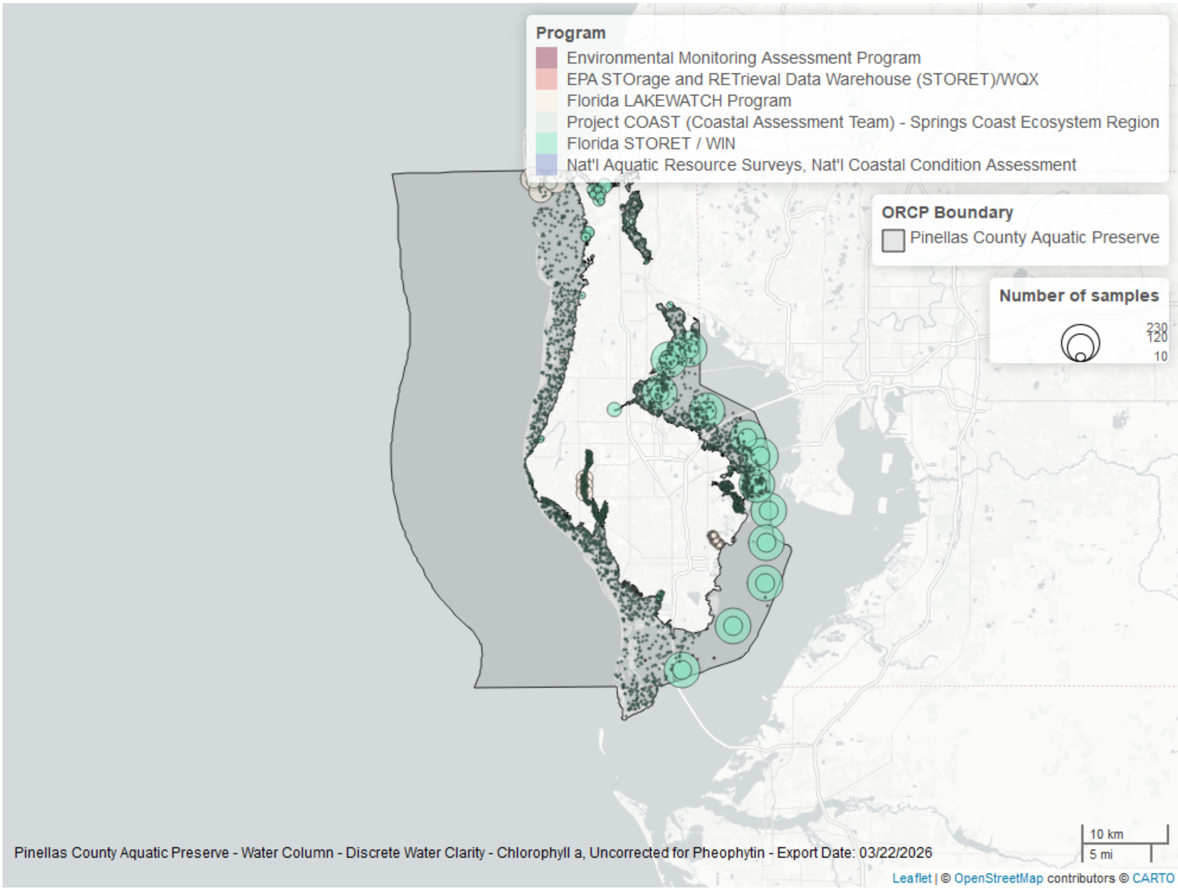


Figure 23: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

### Chlorophyll a, Corrected for Pheophytin - Discrete

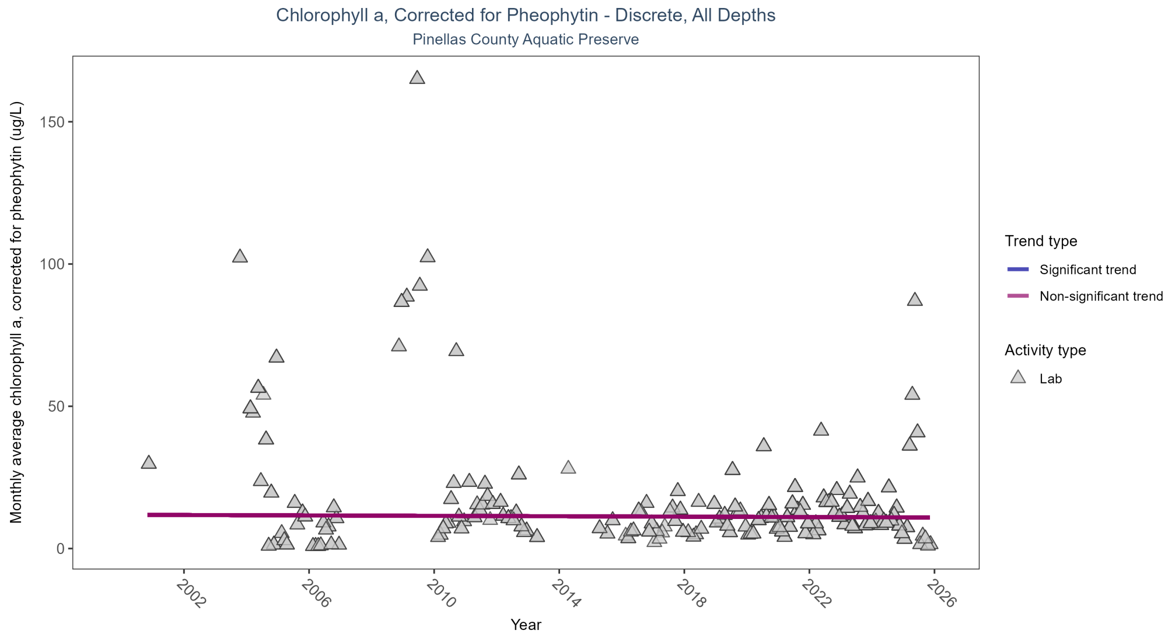


Figure 24: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 13: Seasonal Kendall-Tau Results for - Chlorophyll a, Corrected for Pheophytin

| Activity Type | Statistical Trend | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Lab | No significant trend | 3684 | 23 | 2000 - 2025 | 5.8 | -0.00803 | 11.84857 | -0.03594 | 0.779 |

Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2000 and 2025.

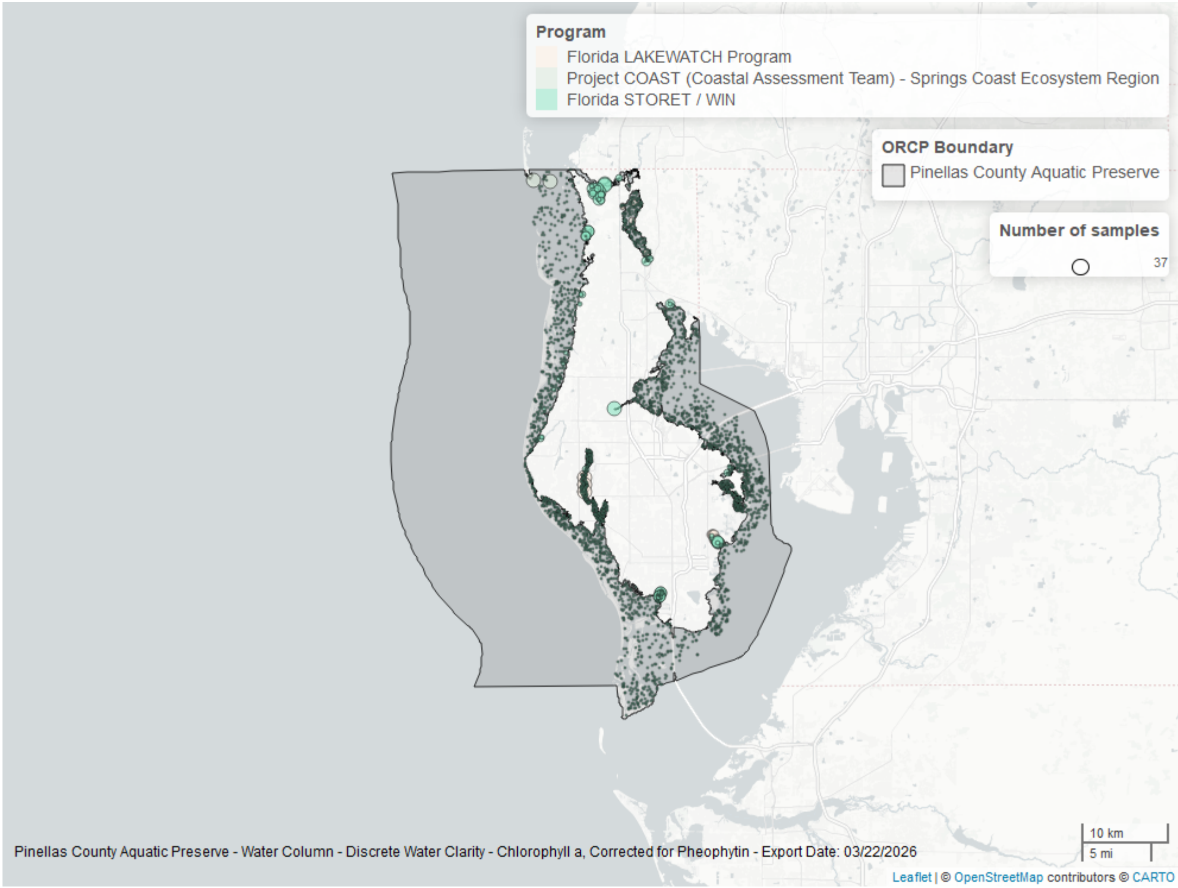


Figure 25: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

### Secchi Depth - Discrete

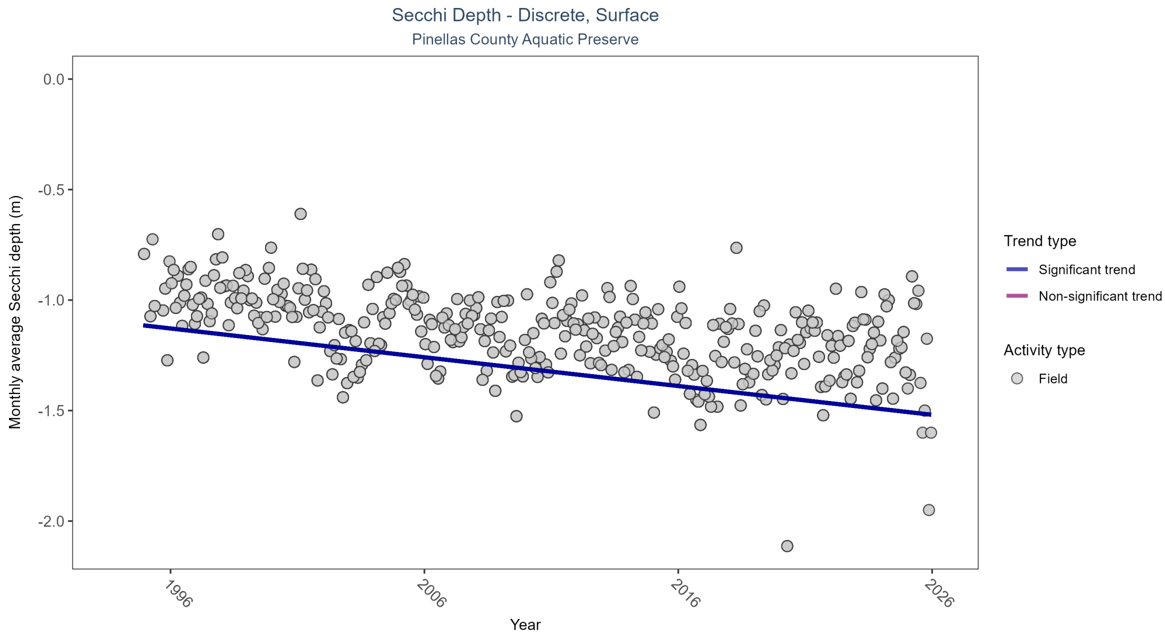


Figure 26: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 14: Seasonal Kendall-Tau Results for - Secchi Depth

| Activity Type | Statistical Trend | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Field | Significantly decreasing trend | 32582 | 32 | 1994 - 2025 | -1.2 | -0.34886 | -1.10239 | -0.01302 | 0 |

Monthly average Secchi depth became deeper by 0.01 m per year, indicating an increase in water clarity.

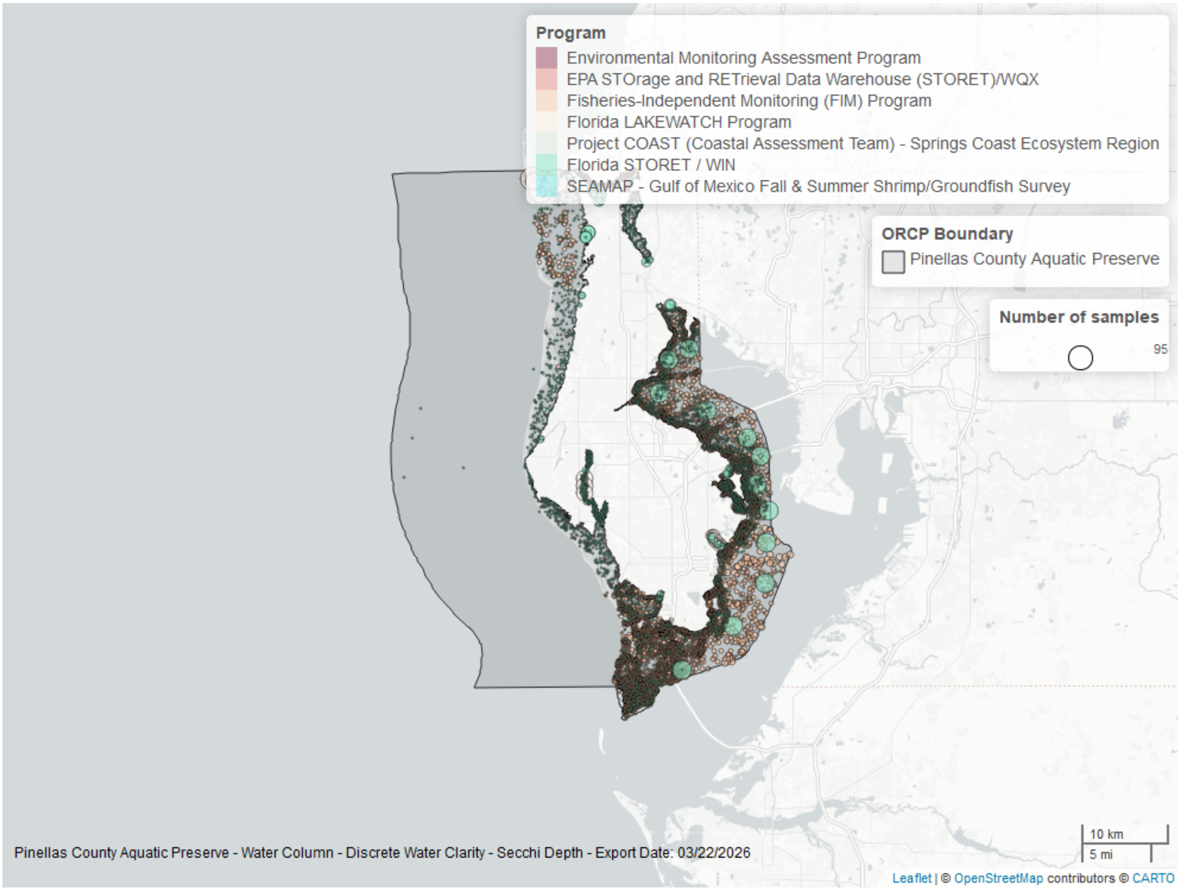


Figure 27: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Colored Dissolved Organic Matter - Discrete

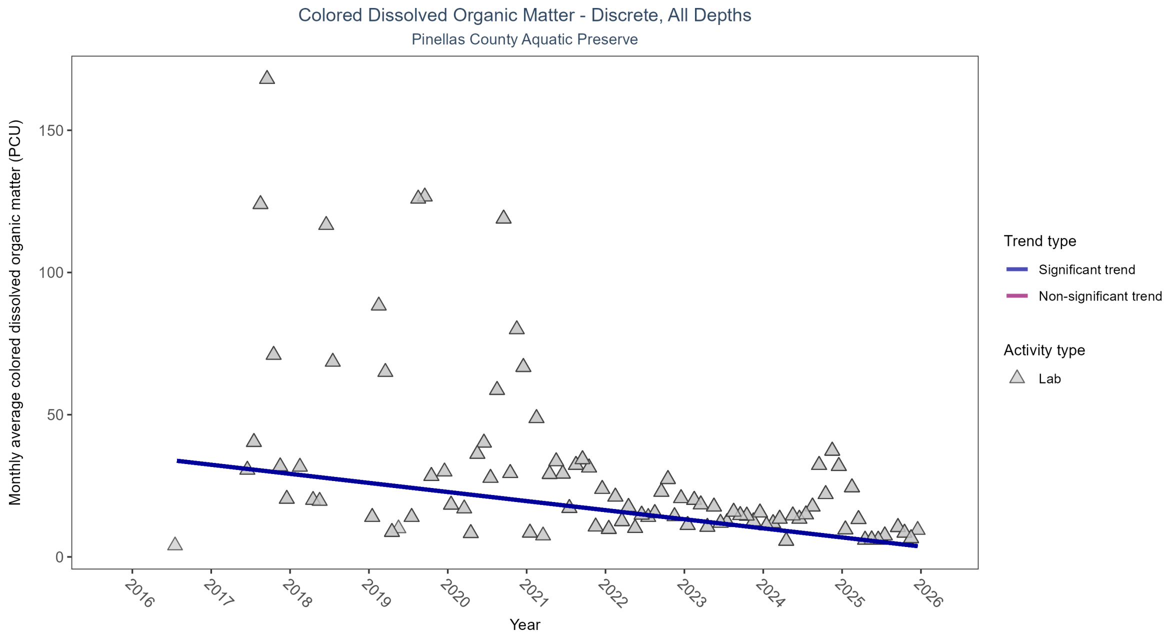


Figure 28: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 15: Seasonal Kendall-Tau Results for - Colored Dissolved Organic Matter

| Activity Type | Statistical Trend | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Lab | Significantly decreasing trend | 1229 | 10 | 2016 - 2025 | 10.4 | -0.4849 | 35.5957 | -3.19339 | 0 |

Monthly average colored dissolved organic matter decreased by 3.19 PCU per year, indicating an increase in water clarity.

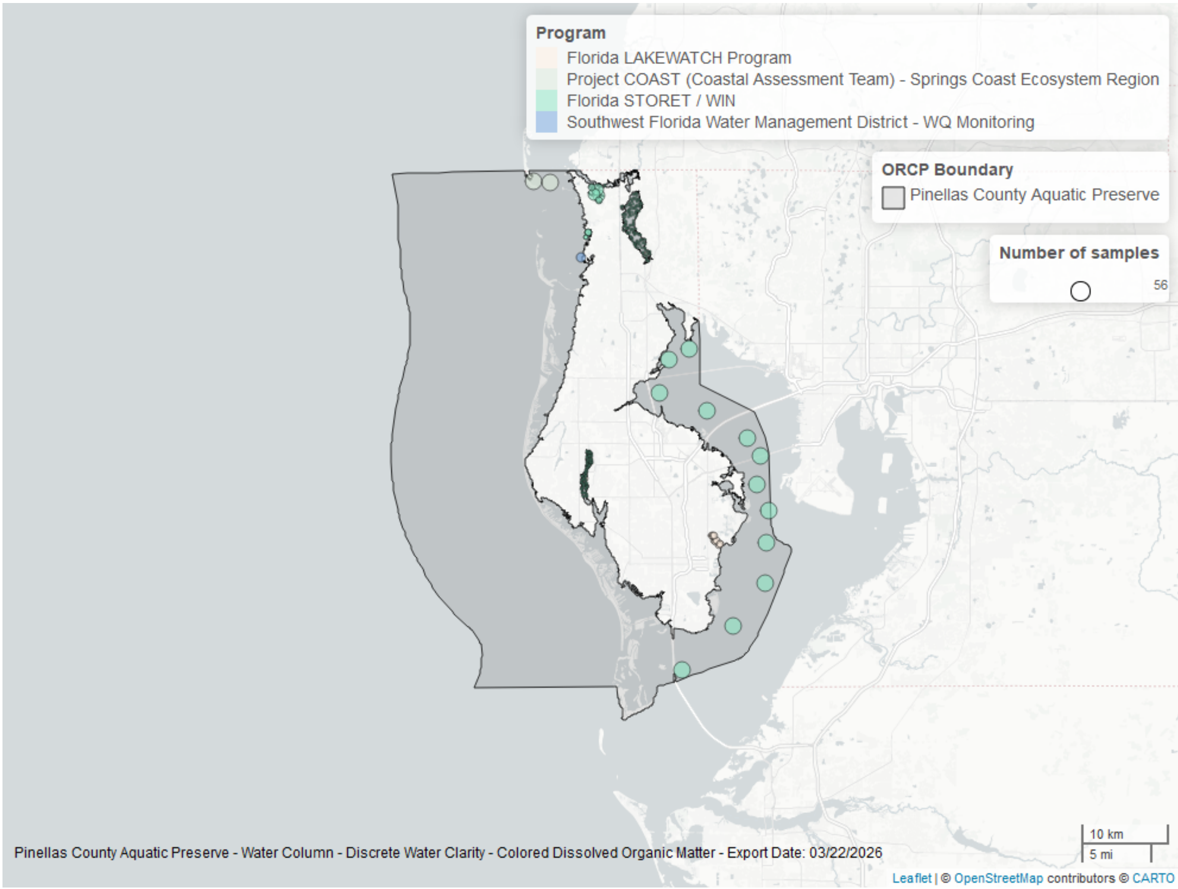


Figure 29: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.